

Madhav K

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EDUCATION

Vellore Institute of Technology (VIT), Chennai

B.Tech in Computer Science & Engineering — AI & Machine Learning Specialization

Expected May 2027

CGPA: 7.94 / 10

EXPERIENCE

Technology Intern

May 2025 – August 2025

Virtusa

- Gained hands-on exposure to production NLP pipelines, observing end-to-end workflows for automated data extraction in an enterprise environment.
- Contributed to data preprocessing and annotation tasks using Python, supporting the team's ML data preparation process.
- Wrote utility scripts for data cleaning and format transformation; reviewed technical documentation to understand pipeline architecture and system design.

Interdisciplinary Technical Content Intern

June 2025 – July 2025

FlyBrand

- Developed predictive and risk analysis models for mechanical systems using synthetic sensor, failure, and maintenance log data.
- Applied regression, classification, and anomaly detection techniques to identify equipment failure patterns and operational risk signals.
- Documented model architectures, methodology, and results as internal technical reference material for the engineering team.

TECHNICAL PROJECTS

EpiGraph-AI — Epidemic Outbreak Prediction | [GitHub](#)

- Designed a hybrid deep learning architecture combining Graph Attention Networks (GAT), LSTMs, and BioBERT to predict dengue fever outbreaks using spatio-temporal clinical, weather, and news data.
- Extracted 768-dim contextual embeddings from health news headlines using BioBERT; compressed to 32 dims via PCA and fused into a unified multi-modal prediction pipeline.
- Modelled inter-district disease spread via graph adjacency matrices with 7-day rolling windows; evaluated on MAE, RMSE, and Outbreak Detection Accuracy using Huber Loss for outlier robustness.

Finstox — Explainable AI Stock Trading Platform | [GitHub](#)

- Designed and integrated an LSTM model for NSE stock price prediction within a FastAPI backend, implementing the full training pipeline using spatio-temporal features.
- Applied LIME to generate per-prediction explanations, improving model interpretability for non-technical users and aligning outputs with XAI principles.
- Contributed to model evaluation metrics and risk analysis components; stack: Python, FastAPI, PyTorch, LSTM, LIME, SHAP.

Predictive Modeling for Peripheral Artery Disease (PAD) Detection | [GitHub](#)

- Engineered 13 clinically-relevant features from Electronic Health Records (EHR) to improve diagnostic signal quality.
- Benchmarked 6 classification algorithms (LightGBM, Random Forest, Logistic Regression, SVM, XGBoost, LSTM); best model achieved AUC of 0.88 on held-out test set.
- Conducted full EDA and model validation pipeline using Python, Pandas, Scikit-learn, and Matplotlib.

LEADERSHIP

Event Lead

July 2025 – Present

Open Source Programming Club, VIT Chennai

- Plan and execute technical workshops, hackathons, and coding events for a community of 600+ students.
- Manage cross-functional student teams across logistics, content, and outreach for each event.

Campus Ambassador

Sep 2025 – Dec 2025

Perplexity AI

- Drove user acquisition campaigns across 3+ universities, achieving registrations beyond campus boundaries including international sign-ups from the US.
- Conducted live demos and outreach initiatives promoting AI-powered research tools to student and faculty communities.

SKILLS

ML & AI: Python, PyTorch, Scikit-learn, TensorFlow, LightGBM, LSTM, Graph Attention Networks (GAT), NLP, EDA

GenAI & LLMs: BioBERT, Transformers (HuggingFace), LangChain, Prompt Engineering, RAG (learning)

Tools & Stack: SQL, Flask, FastAPI, REST APIs, React.js, Git, Pandas, NumPy, Matplotlib, Seaborn

Other: Java, C++, Spatio-temporal Modeling, Anomaly Detection, Risk Analysis, LIME, SHAP